

Indian Institute of Information Technology Vadodara - International Campus Diu



Name: **Saumya Joshi**
Roll no.: **202311039**
Branch: **CSE**
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Subject: **CS 268**

Aim:

To design a data path and control path architecture of repeated adder using Verilog software.

Theory:

Theory: In a complex digital system, the hardware is typically partitioned into two parts:

1. Data Path
2. Control Path

Verilog Code:

datapath.v

```
// Datapath module
module mul_datapath(
    input LdA, LdB, LdP, ClrP, DecB, Clock,
    input [15:0] data_in,
    output eqz
);
    wire [15:0] X, Y, Z, Bout;

    PIP01 A(X, LdA, data_in, Clock);
    PIP02 P(Y, ClrP, LdP, Z, Clock);
    CNTR B(LdB, DecB, Bout, data_in, Clock);
    Add AD(Z, X, Y);
    EQZ comp(Bout, eqz);
endmodule
```

```

endmodule

// Register module for input A
module PIP01(
    output reg [15:0] dout,
    input LdA,
    input [15:0] din,
    input Clock
);
    always @(posedge Clock)
        if (LdA) dout <= din;
endmodule

// Register module for product
module PIP02(
    output reg [15:0] dout,
    input ClrP, LdP,
    input [15:0] din,
    input Clock
);
    always @(posedge Clock)
        if (ClrP) dout <= 16'b0;
        else if (LdP) dout <= din;
endmodule

// Adder module
module Add(
    output reg [15:0] out,
    input [15:0] in1, in2
);
    always @(*)
        out = in1 + in2;
endmodule

// Zero comparison module
module EQZ(

```

```

        input [15:0] Bout,
        output eqz
    );
    assign eqz = (Bout == 0);
endmodule

// Counter module
module CNTR(
    input LdB, Clock, DecB,
    input [15:0] din,
    output reg [15:0] dout
);
    always @(posedge Clock)
        if (LdB) dout <= din;
        else if (DecB) dout <= dout - 1;
endmodule

// Controller module
module controller(
    input Clock, Start, eqz,
    output reg LdA, LdB, LdP, ClrP, DecB, Done
);
    reg [2:0] state;
    parameter s0 = 3'b000, s1 = 3'b001, s2 = 3'b010, s3 = 3'b011, s4 =
3'b100;

    always @(posedge Clock)
        case(state)
            s0: state <= Start ? s1 : s0;
            s1: state <= s2;
            s2: state <= s3;
            s3: state <= eqz ? s4 : s3;
            s4: state <= s4;
            default: state <= s0;
        endcase

```

```

always @(state)
    case(state)
        s0: begin LdA = 0; LdB = 0; LdP = 0; ClrP = 0; DecB = 0;
Done = 0; end
        s1: begin LdA = 1; LdB = 0; LdP = 0; ClrP = 0; DecB = 0;
Done = 0; end
        s2: begin LdA = 0; LdB = 1; LdP = 0; ClrP = 1; DecB = 0;
Done = 0; end
        s3: begin LdA = 0; LdB = 0; LdP = 1; ClrP = 0; DecB = 1;
Done = 0; end
        s4: begin LdA = 0; LdB = 0; LdP = 0; ClrP = 0; DecB = 0;
Done = 1; end
        default: begin LdA = 0; LdB = 0; LdP = 0; ClrP = 0; DecB =
0; Done = 0; end
    endcase
endmodule

```

datapath_tb.v

```

`timescale 1ns/1ps

module datapath_controller_tb;

    reg [15:0] data_in;
    reg Clock, Start;

    wire Done, eqz, LdA, LdB, LdP, ClrP, DecB;

    mul_datapath DP(
        .LdA(LdA),
        .LdB(LdB),
        .LdP(LdP),
        .ClrP(ClrP),
        .DecB(DecB),
        .eqz(eqz),
        .Clock(Clock),
        .data_in(data_in)
    )
endmodule

```

```

);

controller CON(
    .LdA(LdA) ,
    .LdB(LdB) ,
    .LdP(LdP) ,
    .ClrP(ClrP) ,
    .DecB(DecB) ,
    .eqz(eqz) ,
    .Clock(Clock) ,
    .Done(Done) ,
    .Start(Start)
);

initial begin
    Clock = 1'b0;
    Start = 1'b0;
    data_in = 16'b0;
    #3 Start = 1'b1;
end

always #5 Clock = ~Clock;

initial begin
    #17 data_in = 16'd17;
    #10 data_in = 16'd5;
end

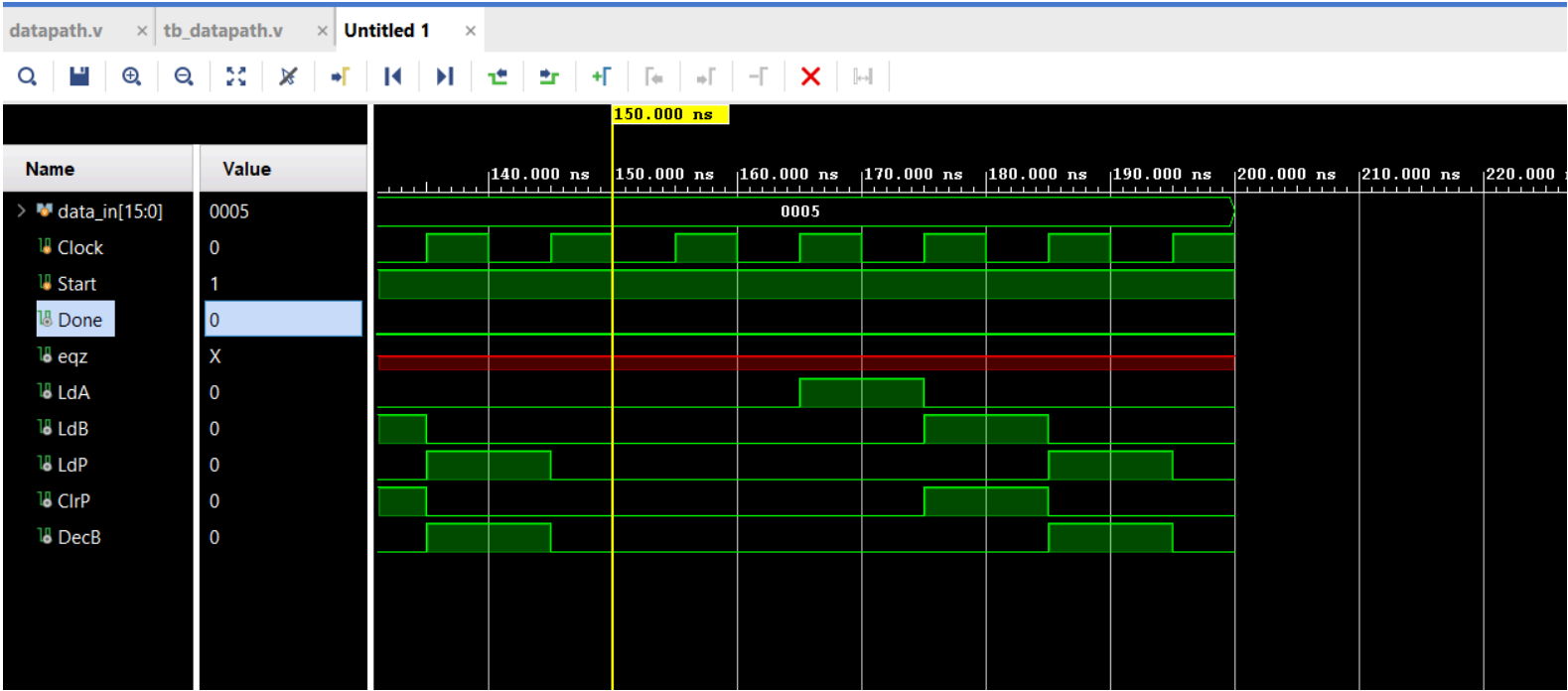
initial begin
    $monitor($time, " Product=%d Done=%b", DP.Y, Done);
    #200 $finish;
end

endmodule

```

Vivado Output:

DIAGRAM:



TCL CONSOLE:

```
# run 1000ns
0 Product=      x Done=x
5 Product=      x Done=0
$finish called at time : 200 ns : File "C:/Users/Saumya Joshi/project_1/project_1.srscs/sources_1/new/tb_datapath.v" Line 47
INFO: [USF-XSim-96] XSim completed. Design snapshot 'datapath_controller_tb_behav' loaded.
INFO: [USF-XSim-97] XSim simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:01 ; elapsed = 00:00:16 . Memory (MB): peak = 1223.176 ; gain = 26.434
```

